Worked solutions

Electronic Circuits I — Practice midterm

Attempt the paper before reading this file. The marking guide is proposed for practice; it is not the lecturer's official rubric. Equivalent valid methods earn credit.

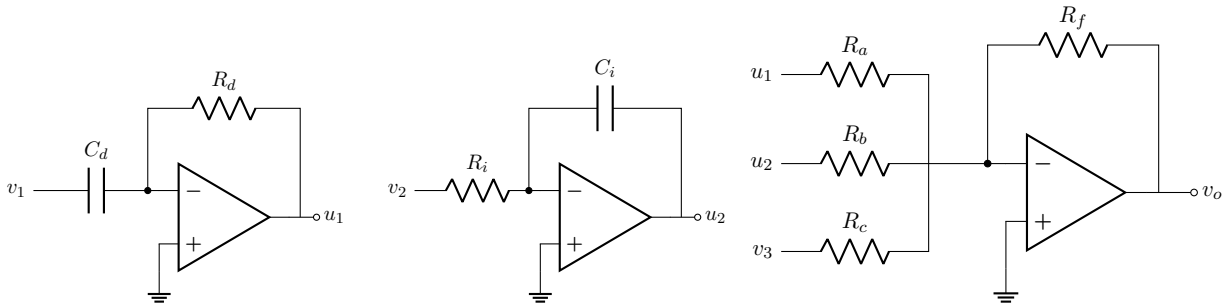
Question 1 [5 marks]

(a) Three-op-amp realization [3]. Use an inverting differentiator, an inverting integrator, and an inverting summer:

$$u_1 = -R_d C_d \frac{dv_1}{dt}, \qquad u_2 = -\frac{1}{R_i C_i}\int_0^t v_2(\tau)d\tau,$$

$$v_o = -\frac{R_f}{R_a}u_1 - \frac{R_f}{R_b}u_2 - \frac{R_f}{R_c}v_3.$$

Choose $R_d = 10\,\text{k}\Omega$, $C_d = 100$ nF, $R_i = 100\,\text{k}\Omega$, $C_i = 100$ nF, $R_f = R_a = R_b = 20\,\text{k}\Omega$, and $R_c = 10\,\text{k}\Omega$. Both processing stages invert; the summer inverts again, producing positive derivative and integral terms and the required negative v_3 term.



Marking: correct topology and signs 1.5; coefficient relations 0.75; consistent values 0.75. All noninverting inputs are grounded. The ideal differentiator is the course model; practical bandwidth limiting is outside this question.

(b) Finite gain [1]. Let $\alpha = R_g/(R_g + R_f) = 0.1$. Then

$$v_o = A_0(v_i - \alpha v_o), \quad A_{CL} = \frac{A_0}{1 + A_0\alpha} = \frac{1000}{101} = 9.90099.$$

The ideal gain is 10, so the reduction is $(10 - 9.90099) \times 100/10 = 0.9901\%$. Award 0.5 for derivation and 0.5 for numerical result.

(c) Log amplifier [1]. For positive input, the feedback transistor carries $I \simeq v_{\text{in}}/R > 0$ and gives a logarithmic relation. Negative input would require reverse feedback current, invalidating this forward-conduction model and the real logarithm. The circuit loses its intended logarithmic feedback behavior; the output may run toward a rail. A bipolar-input log circuit needs a different topology.

Question 2 [7 marks]

(a) R–2R derivation [2]. For code 0010, only b_1 is high. With $b_0 = 0$, the termination and the b_0 branch give $2R \parallel 2R = R$. The path from the b_1 junction toward ground is thus $R + R = 2R$. Its Thevenin equivalent is

$$V_{\text{th},1} = V_{\text{ref}}/2, \qquad R_{\text{th},1} = R.$$

Moving one section right, the zero-valued b_2 branch produces

$$V_{\text{th},2} = V_{\text{ref}}/4, \qquad R_{\text{th},2} = R.$$

The final series R feeds the virtual ground at the MSB junction:

$$I_{0010} = \frac{V_{\text{ref}}/4}{R+R} = \frac{V_{\text{ref}}}{8R}, \qquad v_o(0010) = -\frac{R_f}{R}\frac{V_{\text{ref}}}{8}.$$

By superposition,

$$\boxed{v_o = -\frac{R_f}{R}V_{\text{ref}}\left(\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16}\right) = -\frac{R_f V_{\text{ref}}}{16R}N.}$$

Here $N = 8b_3 + 4b_2 + 2b_1 + b_0$. Award 1.5 for the single-bit derivation and 0.5 for the general expression.

(b) Design [2]. The MSB sees its $2R$ resistor terminating at virtual ground, so $2R = 20\,\text{k}\Omega$ and $R = R_f = 10\,\text{k}\Omega$. At $N = 15$,

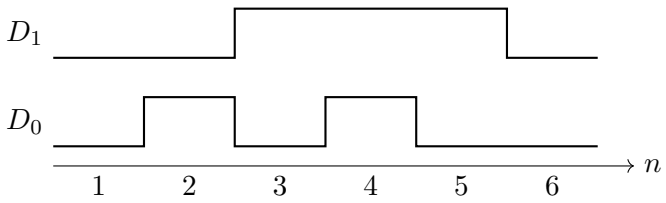
$$-3.75 = -V_{\text{ref}}\frac{15}{16} \quad \Rightarrow \quad V_{\text{ref}} = 4\,\text{V}.$$

R	$2R$	R_f	Logic 0	Logic 1
$10\,\text{k}\Omega$	$20\,\text{k}\Omega$	$10\,\text{k}\Omega$	0 V	4 V

The signed step is $\Delta v_o = -4/16 = -0.25$ V per code. Code 1011 has $N = 11$, giving $v_o = -2.75$ V. Award 0.5 each for resistance choice, reference, step, and code output.

(c) ADC sequence [2].

Sample	1	2	3	4	5	6
v_{in} (V)	0.4	1.2	2.6	3.4	2.2	0.8
$C_3C_2C_1$	000	001	011	111	011	000
D_1D_0	00	01	10	11	10	00



Award 1 for codes and comparator states, 1 for both digital waveforms.

(d) Comparator interpretation [1]. Step size = 4/4 = 1 V; a 3-bit flash ADC requires $2^3 - 1 = 7$ comparators. These op-amps operate open-loop as comparators, with no linear negative-feedback constraint. Thus their inputs generally differ and their outputs select a logic level. Award 0.25, 0.25, and 0.5 respectively.

Question 3 [3 marks]

(a) [1]. A 7812 provides nominally +12 V. At the onset of external-transistor conduction,

$$R_{\text{ext}} = \frac{0.70}{0.40} = 1.75\,\Omega.$$

(b) [1]. $I_L = 12/12 = 1$ A. Under the stated current-sharing model, the external transistor supplies $1 - 0.40 = 0.60$ A, and dissipates

$$P_Q = (18 - 12)(0.60) = 3.60\,\text{W}.$$

This is calculated electrical dissipation, not a complete heat-sink specification.

(c) [1]. The external transistor bypasses the IC's internal limiting path. Add a sense resistor in series with the total output current and a sensing transistor that, at the threshold, reduces pass-device drive through the control/feedback path. The sense resistor is

$$R_s = \frac{0.70}{1.50} = 0.467\,\Omega.$$

Award 0.5 for the protection mechanism and 0.5 for the value. A correct equivalent circuit proposal is acceptable. The sensing arrangement must act on total current, rather than only the IC branch.

Question 4 [4 marks]

(a) Standard astable [2]. $V_L = V_{CC}/3 = 3$ V and $V_H = 2V_{CC}/3 = 6$ V. The capacitor charges through $R_A + R_B$ and discharges through R_B:

$$t_H = (\ln 2)(R_A + R_B)C, \quad t_L = (\ln 2)R_B C.$$

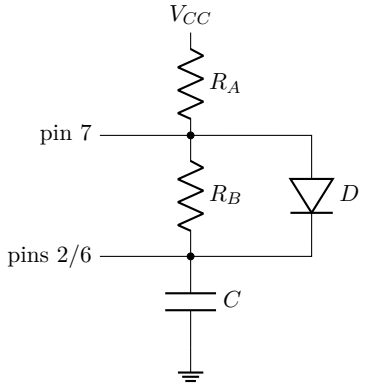
For $T = 1$ ms and $D = 0.75$, $t_H = 0.75$ ms and $t_L = 0.25$ ms:

$$R_B = \frac{0.25\,\text{ms}}{(\ln 2)(100\,\text{nF})} = 3.607\,\text{k}\Omega,$$

$$R_A = \frac{0.75\,\text{ms}}{(\ln 2)(100\,\text{nF})} - R_B = 7.213\,\text{k}\Omega.$$

Award 0.5 for thresholds, 0.5 for equations, and 1 for resistor values.

(b) Modified astable [2]. Place a diode across R_B, with **anode at pin 7 and cathode at the timing-capacitor node (pins 2/6)**. It bypasses R_B during charging and is reverse-biased during discharge.



Now $t_H = (\ln 2)R_A C$ and $t_L = (\ln 2)R_B C$, hence

$$R_A = \frac{0.40\,\text{ms}}{(\ln 2)C} = 5.771\,\text{k}\Omega, \qquad R_B = \frac{0.60\,\text{ms}}{(\ln 2)C} = 8.656\,\text{k}\Omega.$$

Without the diode, $D = (R_A + R_B)/(R_A + 2R_B) > 1/2$ for positive resistors. Award 0.75 for modification/orientation, 1 for values, and 0.25 for the limitation. Values computed using 0.693 or 1.44 are acceptable within rounding.

Question 5. DC and bypassed CE stage [6 of 11 marks]

(a) Physical operation [1]. The forward-biased emitter junction injects electrons into the thin p-type base. Most diffuse across the neutral base without recombining; the reverse-biased collector junction's electric field sweeps them into the collector. Transport through the neutral base is predominantly diffusion. The two junctions share a thin base, so two independent back-to-back diodes do not model this action.

(b) Bias [3]. For DC, all capacitors are open, isolating the source and load:

$$V_{\text{th}} = 8\frac{20}{60+20} = 2\,\text{V}, \qquad R_{\text{th}} = 60\,\text{k}\Omega \parallel 20\,\text{k}\Omega = 15\,\text{k}\Omega.$$

Using $I_E = (\beta+1)I_B$,

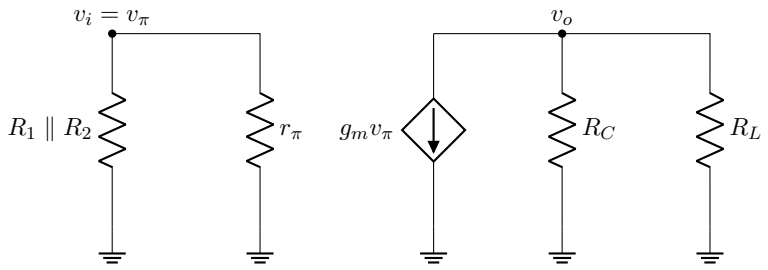
$$I_B = \frac{V_{\text{th}} - V_{BE}}{R_{\text{th}} + (\beta+1)R_E} = \frac{1.3}{116\,\text{k}\Omega} = 11.207\,\mu\text{A}.$$

$$I_C = 1.12069\,\text{mA}, \quad I_E = 1.13190\,\text{mA}, \quad V_E = 1.13190\,\text{V}, \quad V_B = 1.83190\,\text{V}.$$

$$V_C = 8 - I_C R_C = 4.63793\,\text{V}, \qquad \boxed{V_{CE} = 3.50603\,\text{V}.}$$

Since $V_B - V_E = 0.70$ V and $V_C > V_B$, the BE junction is forward biased and the BC junction is reverse biased. The active-region assumption is consistent. Award 0.5 for the Thevenin pair, 1 for currents, 1 for voltages, and 0.5 for the region check.

(c) Midband model and parameters [2]. C_E grounds the emitter for AC, and the DC supply is AC ground. The bias resistors remain in parallel at the input.



Controlled-source current is from collector to emitter (downward).

$$g_m = \frac{I_C}{V_T} = 44.8276\,\text{mS}, \qquad r_\pi = \frac{\beta}{g_m} = 2.23077\,\text{k}\Omega.$$

$$A_v = -g_m(R_C \parallel R_L) = -0.0448276(2000) = -89.655,$$

$$R_{\text{in}} = R_1 \parallel R_2 \parallel r_\pi = 1.94196\,\text{k}\Omega, \qquad R_{\text{out}} = R_C = 3\,\text{k}\Omega.$$

Award 0.5 for the model, 0.5 for g_m, r_π, 0.5 for gain and 0.5 for both impedances. If a finite Early voltage were specified, r_o would be included; it is infinite here by assumption.

Question 5. Degeneration, follower and CB [5 of 11 marks]

(d) Remove the bypass capacitor [2]. Write $v_i = i_b r_\pi + i_e R_E$ with $i_e = (\beta+1)i_b$. The collector signal is $v_o = -\beta i_b (R_C \parallel R_L)$. Therefore,

$$\boxed{A_v = -\frac{\beta(R_C \parallel R_L)}{r_\pi + (\beta+1)R_E}} = -\frac{100(2000)}{2230.77 + 101(1000)} = -1.93741.$$

$$\boxed{R_{\text{in}} = R_1 \parallel R_2 \parallel [r_\pi + (\beta+1)R_E]} = 13.0969\,\text{k}\Omega.$$

The DC operating point is unchanged because the bypass capacitor was already open in DC. Degeneration decreases gain magnitude and increases input resistance. Award 1 for gain derivation/result, 0.75 for input resistance, and 0.25 for the DC conclusion.

(e) Emitter follower [2]. Here $r_\pi = 100/0.040 = 2.5\,\text{k}\Omega$ and the loaded emitter resistance is $R'_E = 1\,\text{k}\Omega \parallel 1\,\text{k}\Omega = 500\,\Omega$. Since the ideal base bias current source has infinite small-signal impedance,

$$R_{\text{in,base}} = r_\pi + (\beta+1)R'_E = 2500 + 101(500) = 53\,\text{k}\Omega,$$

$$\boxed{\frac{v_e}{v_b} = \frac{(\beta+1)R'_E}{r_\pi + (\beta+1)R'_E} = \frac{50500}{53000} = 0.95283.}$$

This is the requested *base-to-emitter* gain. The source resistance is not included in that ratio. If one instead wanted source-to-output gain, it would be $0.95283 \times 53/(53+3) = 0.90179$.

For output resistance, remove R_L, set the signal source to zero, and retain R_s from base to ground:

$$\boxed{R_{\text{out}} = R_E \parallel \frac{r_\pi + R_s}{\beta+1}} = 1000 \parallel \frac{2500+3000}{101} = 51.643\,\Omega.$$

Award 0.5 for input resistance, 0.5 for gain, and 1 for output resistance including the source resistance and correct load removal.

(f) Common-base stage [1]. $g_m = 1\,\text{mA}/25\,\text{mV} = 40$ mS and $r_\pi = 2.5\,\text{k}\Omega$. The base is AC ground, so $v_\pi = -v_e$. KCL at the collector gives $v_c = +g_m R'_C v_e$:

$$\boxed{A_v = +0.04(2000) = +80,} \qquad \boxed{R_{\text{in,e}} = \frac{1}{g_m + 1/r_\pi} = \frac{r_\pi}{\beta+1} = 24.7525\,\Omega.}$$

The large-β approximation is $1/g_m = 25\,\Omega$. Award 0.5 for each quantity; accept the approximation if stated.

Mark check: $5 + 7 + 3 + 4 + 11 = 30$. Carry a consistent earlier numerical error forward without repeatedly penalizing it. Deduct separately for a new conceptual error (for example, neglecting the load in the gain or imposing a virtual short on a comparator).